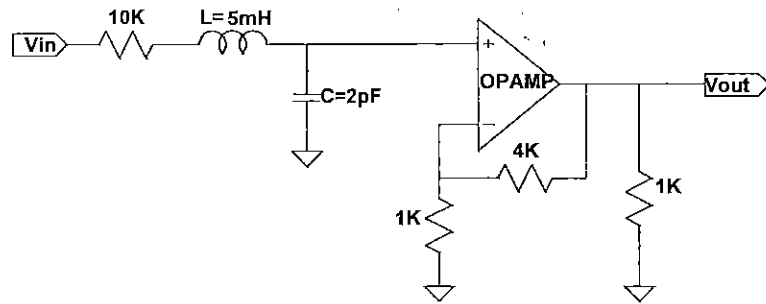


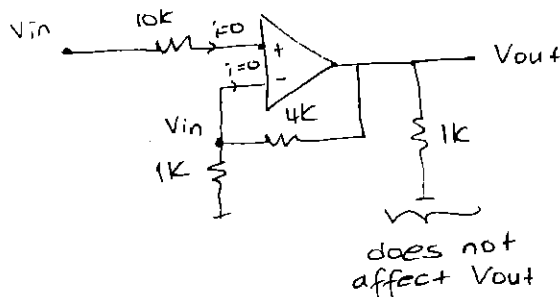
Question 1 [25 Points]:



For the circuit above, assume that the OPAMP is not saturated.

(a) [7 points] Find V_{out} if $V_{in} = 3$ Volts (DC).

For DC inputs, L is short circuit, C is open circuit.



$$V_{out} \cdot \frac{1K}{1K + 4K} = V_{in}$$

$$V_{out} = 5 \cdot V_{in} = \boxed{15 \text{ Volts (DC)}}$$

(b) [5 points] What is the resonance frequency (f_0) of the LC circuit in units of Hz?

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 2 \times 10^{-12}}} = \frac{10^7}{2\pi} = \boxed{1.59 \text{ MHz}}$$

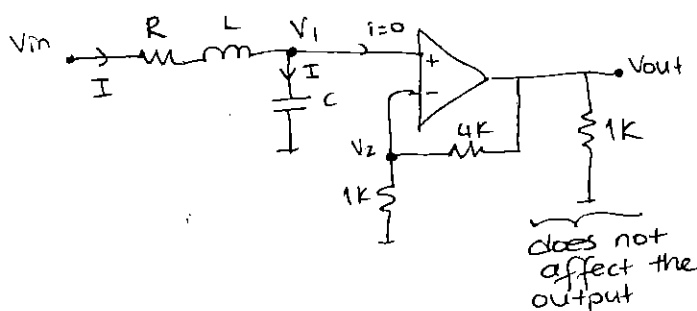
(c) [5 points] What is the Q of the LC tuned circuit? What is its 3-dB bandwidth (in Hz)?

For series LC tuned circuit, $Q = \frac{\omega L}{R} = \frac{1}{\omega_0 RC}$

$$Q = \frac{\omega_0 L}{R} = \frac{10^7 \cdot 5 \times 10^{-3}}{10 \times 10^3} = \boxed{5}$$

$$Q = \frac{f_0}{\Delta f} \Rightarrow \Delta f = \frac{f_0}{Q} = \frac{1.59 \times 10^6}{5} = \boxed{318.3 \text{ kHz}}$$

(d) [8 points] Find V_{out} if $V_{in} = 2 \cos(\omega_0 t + \pi/3)$ Volts, with ω_0 at resonance frequency.



$$I = \frac{V_{in}}{R} \quad (\text{LC appear like short circuit at } \omega_0)$$

$$V_1 = I \cdot \frac{1}{j\omega C} = \frac{V_{in}}{j\omega RC} = -jQ \cdot V_{in}$$

$$V_2 = V_{out} \cdot \frac{1K}{1K + 4K} = \frac{V_{out}}{5}$$

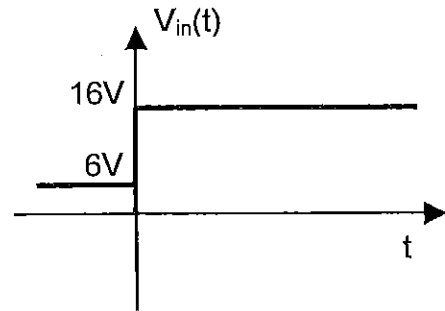
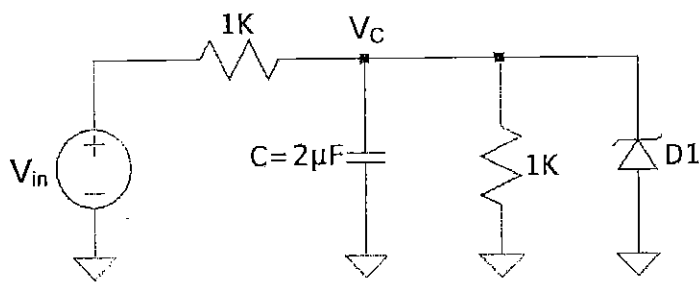
$$V_1 = V_2 \Rightarrow V_{out} = -j5Q \cdot V_{in}$$

$$V_{out} = -j25 V_{in} = 25 V_{in} \cdot e^{-j\pi/2}$$

$$\text{So, } V_{out} = 50 \cdot \cos(\omega_0 t + \pi/3 - \pi/2)$$

$$V_{out} = \boxed{50 \cdot \cos(\omega_0 t - \pi/6) \text{ Volts}}$$

Question 2 [25 points]:



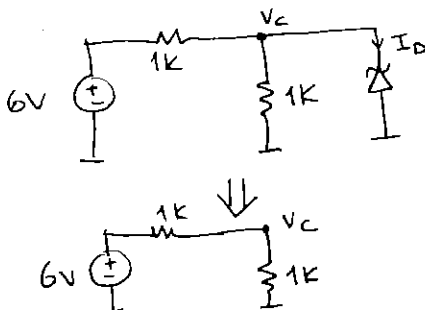
In the circuit above, the breakdown voltage for the zener diode D1 is $V_Z = 6V$.

$V_{in}(t)$ is as given above.

(a) [7 points] What is the current through the zener diode at $t=0^-$ (right before $t=0$)?

What is V_C at $t=0^-$?

at $t=0^-$, V_{in} has been on for a long time, so capacitor is open circuit.

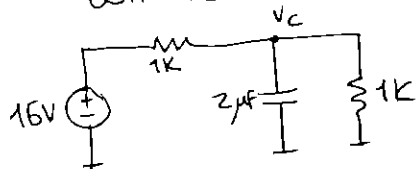


Zener diode is OFF, $I_D(0^-) = 0$

$$\text{Hence, } V_C(0^-) = 6 \cdot \frac{1K}{1K+1K} = \boxed{3V}$$

(b) Find and plot $V_C(t)$ for $-\infty < t < \infty$. Mark all important voltage levels and time points on the axis given below.

$V_C(t)$ cannot change instantaneously. So, Zener diode will remain OFF initially.



$$V_{init} = 3V$$

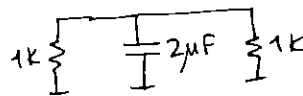
$$V_{final} = \frac{16V}{2} = 8V$$

(calculate by setting capacitor to open circuit)

$$\tau = RC = (0.5 \times 10^3 \times 2 \times 10^{-6})$$

$$\boxed{\tau = 1ms}$$

* Kill all sources to find τ :



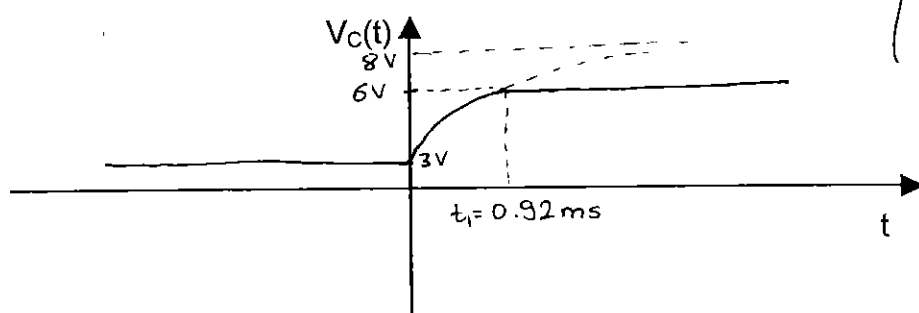
$$\text{So, } V_C(t) = V_{final} + (V_{init} - V_{final})e^{-t/\tau} = 8 - 5e^{-t/\tau}$$

However, once $V_C(t)$ reaches 6V, Zener diode goes into breakdown mode, and sets V_C to 6V. This happens when:

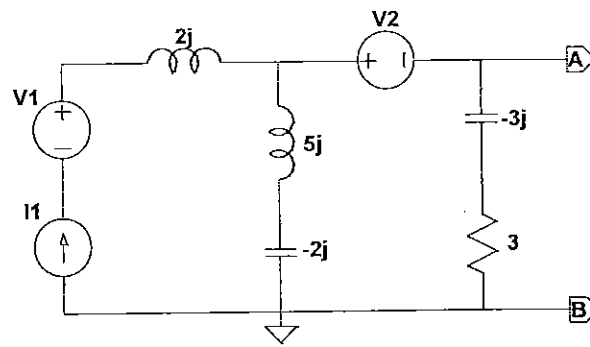
$$V_C(t) = 8 - 5e^{-t_1/\tau} = 6 \Rightarrow t_1 \approx 0.92ms$$

$$\text{So, } V_C(t) = \begin{cases} 3V, & t < 0 \\ 8 - 5e^{-t/\tau}, & 0 \leq t \leq 0.92ms \\ 6V, & t > 0.92ms \end{cases}$$

where $\tau = 1ms$



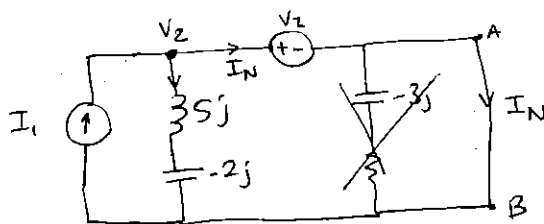
Question 3 [25 points]:



In the circuit above, $I_1(t) = 5 \cos(\omega t)$, $V_1(t) = 8 \sin(\omega t - \pi/4)$, $V_2(t) = 6 \sin(\omega t)$, and all of the impedances are provided in units of Ohms at frequency ω . $\cos(\omega t - \pi/2)$

(a) [8 points] Find the Norton current $I_N(t)$ in time domain, across nodes A&B.

V_1 and $2j$ have no effect on the current on that branch.



$$I_1 = 5 ; V_2 = 6e^{-j\pi/2} = -6j$$

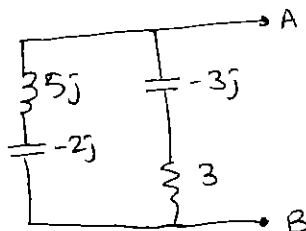
$$* I_N + \frac{V_2}{5j - 2j} = I_1$$

$$I_N = I_1 - \frac{V_2}{3j} = 5 + \frac{6j}{3j} = 7$$

$$I_N(t) = 7 \cos(\omega t)$$

(b) [9 points] Find the equivalent impedance (Z_{eq}) across nodes A&B at frequency ω .

Kill all sources.



$$Z_{eq} = (3 - 3j) \parallel (5j - 2j) = (3 - 3j) \parallel 3j$$

$$= \frac{(3 - 3j) \cdot 3j}{3 - 3j + 3j} = (3 - 3j) \cdot j$$

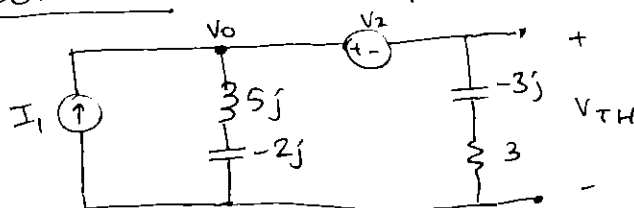
$$\boxed{Z_{eq} = 3 + 3j}$$

(c) [8 points] Find the Thevenin voltage $V_{TH}(t)$ in time domain, across nodes A&B.

Solution 1: $V_{TH} = I_N \cdot Z_{eq} = 7 \cdot (3 + 3j) = 21\sqrt{2} e^{j\pi/4}$

So, $V_{TH}(t) = 21\sqrt{2} \cos(\omega t + \pi/4) \approx \boxed{29.7 \cos(\omega t + \pi/4)}$

Solution 2: Directly find V_{TH} .



$$\frac{V_0}{5j - 2j} + \frac{V_0 - V_2}{3 - 3j} = I_1$$

$$\frac{V_0}{3j} + \frac{V_0 + 6j}{3 - 3j} = 5$$

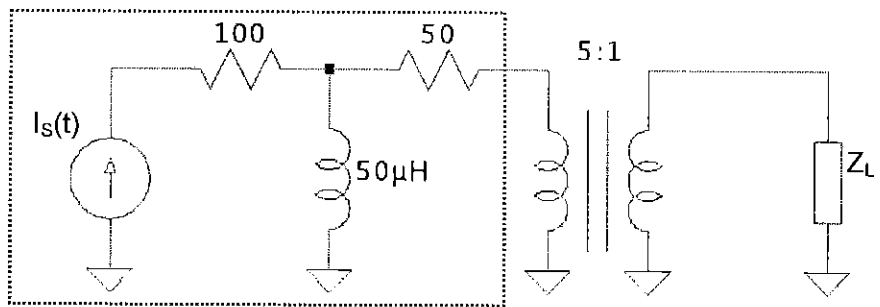
$$V_0 - 6 = 5 \cdot 3 \cdot (1 - j) \cdot j$$

$$V_0 = 21 + 15j$$

$$V_{TH} = V_0 - V_2 = 21 + 21j = 21\sqrt{2} e^{j\pi/4}$$

$$\boxed{V_{TH} \approx 29.7 \cos(\omega t + \pi/4)}$$

Question 4 [25 points]:

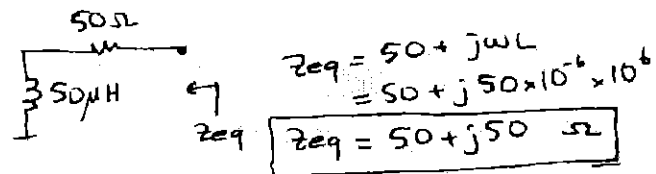


In the circuit above, the transformer is an ideal transformer with a turns ratio of 5:1. The current source is $I_s(t) = 3 \cos(\omega_0 t + \pi/6)$ mA, where $\omega_0 = 10^6$ rps.

(a) [10 points] Find the Thevenin equivalent of the circuit inside the dashed box.

$$I_s = 3 e^{j\pi/6}$$

* Find Z_{eq} : kill all sources.

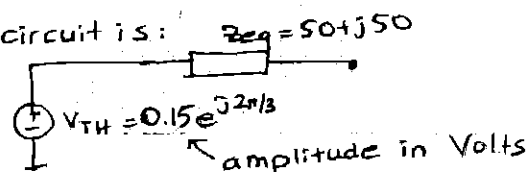


* Find V_{TH} :



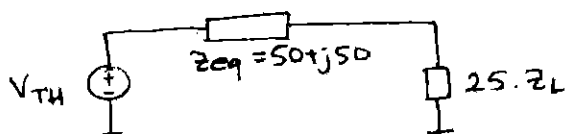
$$V_{TH} = I_s \cdot j\omega L = 3 e^{j\pi/6} \cdot j50 = 150 e^{j(\pi/6 + \pi/2)} = 150 e^{j2\pi/3} \text{ mV} = 0.15 e^{j2\pi/3} \text{ Volts}$$

So, the Thevenin eq. circuit is:



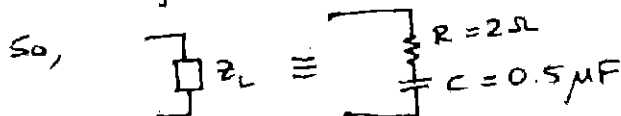
(b) [10 points] Find Z_L that maximizes the power delivered to the load. How would you implement this load? Draw Z_L and specify R, L or C values that you would use.

The source sees the load as $\left(\frac{n_1}{n_2}\right)^2 Z_L = 25 Z_L$

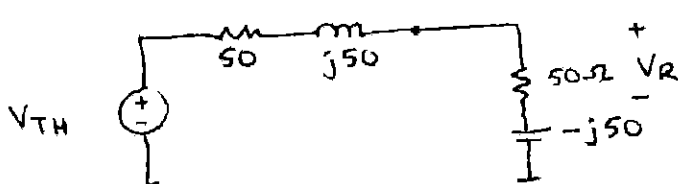


* To maximize power delivered, we need: $25 Z_L = Z_{eq}^*$

$$* -j2 = \frac{1}{j\omega C} \Rightarrow C = \frac{1}{2\omega} = 0.5 \mu\text{F}$$



(c) [5 points] Calculate this maximum power dissipated in the load.



← only the resistor dissipates power.

$$V_R = \frac{V_{TH} \cdot 50}{50 + j50 + 50 - j50} = \frac{V_{TH}}{2}$$

$$P = \frac{1}{2} \frac{|V_R|^2}{R} = \frac{1}{2} \frac{(0.075)^2}{50} = 5.625 \times 10^{-5} \text{ Watts} = 56.25 \mu\text{W}$$

to convert to rms